SEFU HMIS — Functional & System Requirements Specification (FRS / SRS)

Kenya Digital Health Agency (DHA) Certification Compliance Document

Category: System Documentation — Section D: Technical & Functional Specification

System Classification: Health Information System (HIS) & Electronic Medical Record (EMR)

Applicable Regulations: Kenya Health Information Management Procedures Regulations 2025 (Form HMIS 4), Kenya Digital Health Act 2023, Kenya Data Protection Act 2019

■ 1. Document Control & System Release Metadata

Attribute	Specification
Document Title	Functional Requirements Specification (FRS) & System Requirements Specification (SRS)
System Name	SEFU Hospital Management Information System (SEFU HMIS)
System Version	Release 5.0 (Build 2026-Kenya-DHA)
Document Version	1.0 (Production Release / Statutory Certification Filing)
Document Owner	Chief Technology Officer & Lead Solutions Architect
Target Reviewers	Kenya Digital Health Agency (DHA) Evaluation Team & Hospital Technical Committees
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Classification	Public Release Technical Document — Kenya Statutory Filing

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■ 2. System Overview & Executive Summary

The **SEFU Hospital Management Information System (SEFU HMIS)** is an enterprise, modular digital health solution designed to manage inpatient, outpatient, diagnostic, pharmaceutical, surgical, and financial healthcare workflows.

Built on an optimized **Model-View-Controller (MVC)** architecture, SEFU HMIS provides digital medical records, real-time clinical decision documentation, automated itemized billing, multi-factor authentication (MFA), immutable audit trails, and granular Role-Based Access Control (RBAC).

■■ 3. High-Level System Architecture

SEFU HMIS utilizes a robust 4-tier modular architecture designed for high availability, security isolation, and rapid data throughput:

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flowchart TD
subgraph ClientTier ["1. Presentation / Client Tier"]
UI1["Clinical Workstation Browser\n(Chrome, Firefox, Edge)"]
UI2["Nurse Station Tablet\n(Mobile Safari, Android)"]
UI3["Patient Portal / Online Services\n(Public Web / Patient Dashboard)"]
end

subgraph SecurityTier ["2. Gateway & Security Enforcement Tier"]
GW1["TLS 1.2 / 1.3 Termination & Reverse Proxy"]
GW2["CodeIgniter Security Filters (CSRF, XSS Clean, WAF)"]
GW3["Authentication & MFA Gatekeeper (Email OTP Challenge)"]
end

subgraph AppTier ["3. Application / Business Logic Tier (MVC)"]
Ctrl["Controllers (Admin, Patient, Audit, Specialty Modules)"]
CoreLib["Core Libraries (Customlib, RBAC, Mailer, Datatables)"]
Models["Data Models (Patient, Pharmacy, Billing, MFA, Audit)"]
end

subgraph DataTier ["4. Persistent Data & Storage Tier"]
DB[("MariaDB 10.4+ / MySQL Database\n(Transactional Clinical Storage)")]
AuditLog[("Audit Trails & Userlog Engine\n(Activity & Authentication History)")]
MediaStorage["Encrypted Document & Media Vault\n(/uploads/ Scans, Lab Slips)"]
BackupStorage["3-2-1 Automated Backup Repository\n(/backup/ + Offsite Cloud Sync)"]
end

ClientTier --> SecurityTier
SecurityTier --> AppTier
AppTier --> DataTier
```

■ 4. Technical System Requirements Specification (SRS)

4.1 Server Infrastructure Requirements (Production Hospital Deployment)

Hardware / Resource	Minimum Baseline (Level 3-4 Hospital)	Recommended Enterprise (Level 5-6 Hospital)
Processor (CPU)	4 Cores (2.4 GHz x86-64)	8 to 16 Cores (Intel Xeon / AMD EPYC)
System Memory (RAM)	8 GB DDR4	16 GB to 32 GB DDR4/DDR5
Storage (Database)	250 GB NVMe SSD (RAID 1 Mirror)	1 TB NVMe SSD (RAID 10 Redundant Array)
Storage (Media / Scans)	500 GB Storage	2 TB Dedicated Storage Server / NAS
Network Interface	1 Gbps Dedicated Ethernet	Dual 10 Gbps Redundant NICs (Bonded)
Power Infrastructure	Online Double-Conversion UPS (45 min)	Dual UPS + Automated Diesel Generator (< 12s ATS)

4.2 Software Technology Stack

Stack Component	Specified Technology	Purpose & Capabilities
Operating System	Ubuntu 22.04 LTS / Windows Server 2022	Enterprise host server environment
Web Server	Apache 2.4 / Nginx (HTTP/2 enabled)	Request handling, virtual hosts, SSL termination
Application Runtime	PHP 8.1 / 8.2 (with FPM)	Server-side execution engine
Core Framework	CodeIgniter 3.1.13 (Hardened)	High-throughput MVC application framework
Database Engine	MariaDB 10.4+ / MySQL 8.0+	Relational, ACID-compliant health database
Cryptographic Engine	OpenSSL (AES-256 / SHA-256)	In-transit and at-rest cryptographic validation
Client Interface	HTML5, Bootstrap 3.3.7, jQuery, CSS3	Responsive interface optimized for desktops and tablets

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■■ 5. Functional Requirements Specification (FRS) by Module

5.1 Patient Identification & Master Patient Index (MPI)

- **FR-01 (Registration):** Capture patient name, national/alien ID, gender, DOB, marital status, phone, email, and address.
- **FR-02 (Unique Identifier):** System automatically generates an immutable alphanumeric patient ID.
- **FR-03 (Search & Demographics):** Fast indexed search across patient ID, mobile number, national ID, and patient name.

5.2 Outpatient Department (OPD)

- **FR-04 (Visit Intake):** Create new clinical visits, assign consulting physician, and record vitals (BP, pulse, height, weight, BMI).
- **FR-05 (Clinical Notes):** Record presenting symptoms, clinical observations, physical exam, diagnosis, and ICD-10 codes.
- **FR-06 (Prescription Writing):** Electronic prescribing linked to hospital pharmacy stock, with dosage, interval, and instructions.

5.3 Inpatient Department (IPD)

- **FR-07 (Admission & Bed Management):** Allot inpatient beds by ward category (ICU, General, Private, Maternity).
- **FR-08 (Inpatient Charting):** Consultant visit log, medication administration record (MAR), nurse notes, and treatment progress.
- **FR-09 (Discharge Summaries):** Comprehensive medical discharge summary generation including treatment summary and follow-up advice.

5.4 Specialty Clinical Modules

- **FR-10 (Physiotherapy Module):** Specialized assessment of muscular/skeletal pain, pain scale (1–10), anatomical location, therapy session duration, specialized equipment, session progress, and automated billing voucher (**PHYB**).
- **FR-11 (Pediatric Module):** Pediatric intake capturing exact birth date, day/month age calculation, guardian details, birth weight, delivery mode, immunization tracking, pediatric consultations, and automated billing voucher (**PEDB**).
- **FR-12 (Gynaecology Module):** Women's health clinical documentation capturing Last Menstrual Period (LMP), Expected Date of Delivery (EDD), Gravida/Para/Abortion (GPA) obstetric history, antenatal visits, and automated billing voucher (**GYNB**).

5.5 Diagnostics (Laboratory & Radiology)

- **FR-13 (Pathology):** Test ordering, barcode sample tracking, reference range validation, test parameter entry, and pathologist approval.
- **FR-14 (Radiology):** Diagnostic imaging requests (X-Ray, Ultrasound, CT, MRI), clinical indications, and radiologist reporting.

5.6 Pharmacy & Medication Management

- **FR-15 (Medicine Inventory):** Medicine category, supplier, batch number, manufacturing date, and expiry date alerts.
- **FR-16 (Dispensing & Invoicing):** Automated reduction of stock upon dispensing, POS billing, and expiry warnings.

5.7 Hospital Billing, Cashier & TPA Insurance

- **FR-17 (Itemized Billing):** Automated aggregation of bed charges, consultations, procedures, lab tests, and medications.
- **FR-18 (Insurance / TPA Mapping):** Policy numbers, approval codes, co-pays, and insurance claim voucher exports.
- **FR-19 (Cashier Receipts):** Official receipts with Kenya Shilling (KES) currency and hospital branding.

5.8 Information Security, MFA & Auditing

- **FR-20 (Password Security):** Enforce 8+ character password complexity with Bcrypt hashing (`$2y$10$...`).
- **FR-21 (Multi-Factor Authentication):** Enforce 6-digit Email OTP with 10-minute validity and 5-attempt brute-force lockout for staff.
- **FR-22 (Audit Logging):** Immutable recording of every Insert, Update, and Delete action in `logs` table with staff ID and role attribution.
- **FR-23 (Session Management):** Automatic 2-hour idle timeout, 5-minute session ID rotation, and `HttpOnly`/`SameSite` cookie flags.

■ 6. Role-Based Access Control (RBAC) Matrix

SEFU HMIS enforces 13 specialized roles, each configured with granular CRUD permissions:

Role ID	Role Name	System Access Scope	Default Permissions
1	**Admin**	Full operational administration	Full CRUD on all modules, Reports, Staff management
2	**Accountant**	Financial and cashier management	Full access to Billing, Income, Expenses, Financial Reports
3	**Doctor**	Inpatient and Outpatient clinical care	View/Edit Patients, OPD/IPD consultations, Prescriptions
4	**Pharmacist**	Pharmacy inventory & dispensing	Full access to Pharmacy, Medicine stock, Medicine billing
5	**Pathologist**	Laboratory diagnostics	Full access to Pathology tests, parameters, and lab results
6	**Radiologist**	Medical imaging	Full access to Radiology orders, scans, and reports
7	**Super Admin**	Complete root system authority	Root privileges across entire database, server, and settings
8	**Receptionist**	Front office, appointments, triage	Patient registration, Appointments, Visitor log, Bed view
9	**Nurse**	Ward management & patient care	Vital signs, Medication administration, Nurse notes
10	**Physiotherapist**	Physical therapy specialty	Physiotherapy assessment, sessions, and treatment notes
11	**Pediatrician**	Child healthcare specialty	Pediatric growth tracking, immunizations, and clinical notes
12	**Gynaecologist**	Women's health specialty	Antenatal records, LMP/EDD tracking, obstetric consultations

13	**Human Resource**	Staff management & system auditing	Staff directory, Payroll, Attendance, **Audit Trails & User Logs**

Approved by:

Chief Technology Officer & Lead Solutions Architect

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